

EMI and EMC

Basics of EMI and EMC

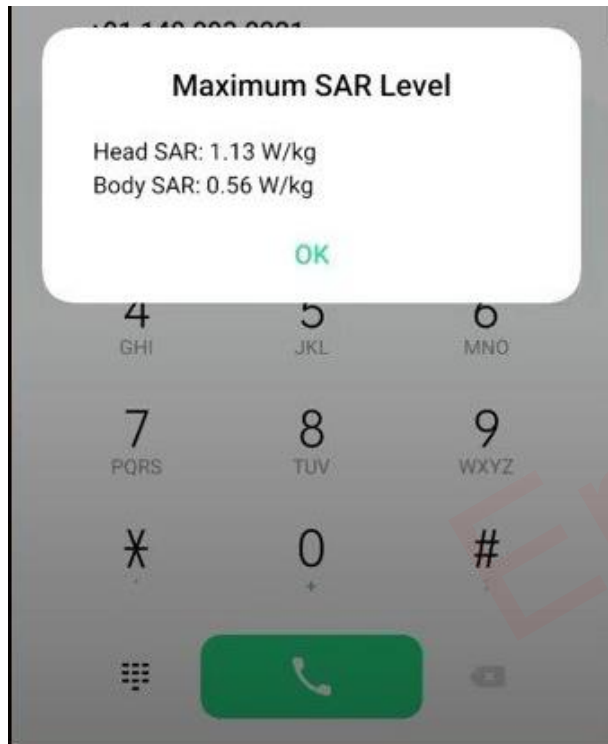
- ❑ EMI {Electromagnetic Interference} – It explains the generation of EM Interference.
- ❑ Sources of EMI
 - Natural Sources: Lightning, Solar Flares etc.
 - Man-made Sources: Motors, RF Transmitters, Circuits etc.
- ❑ Types of EMI
 - Conducted EMI: Through Power Lines, Signal Lines.
 - Radiated EMI: Through Air {EM waves}.
- ❑ EMC {Electromagnetic Compatibility} – EMC controls EMI to work with the given device under a given environment.
- ❑ Goal of EMC
 - Emission Control of EM waves.
 - Immunity Control against EMI.

EMC Classes

❑ EMC explains standards, Up to what EMI, given device allowed to function properly.

❑ EMC Classes are:

- **Emission:** It explains the radiation Class of a given device.
- **Susceptibility:** It explains the functioning of a given device under RFI.
- **Coupling:** It explains the Intra and Inter-Coupling of RF signals.



After giving command

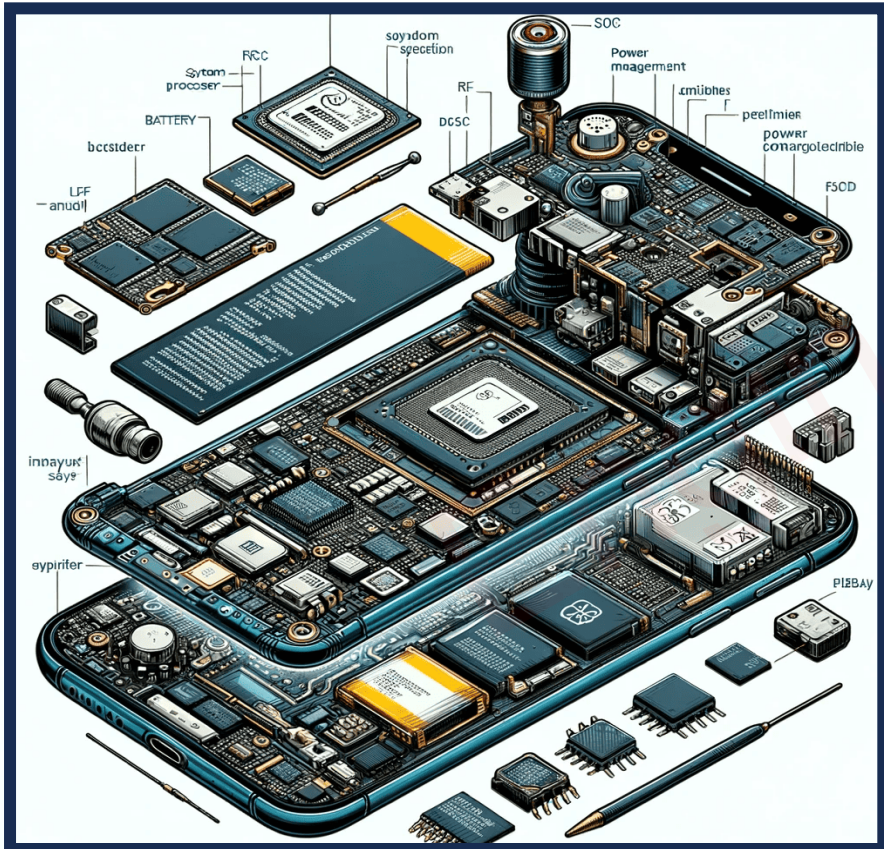
***#07#**

EMI Classes

❑ EMI explains Electromagnetic Interference / RF Interference by a given device.

❑ EMI Classes are:

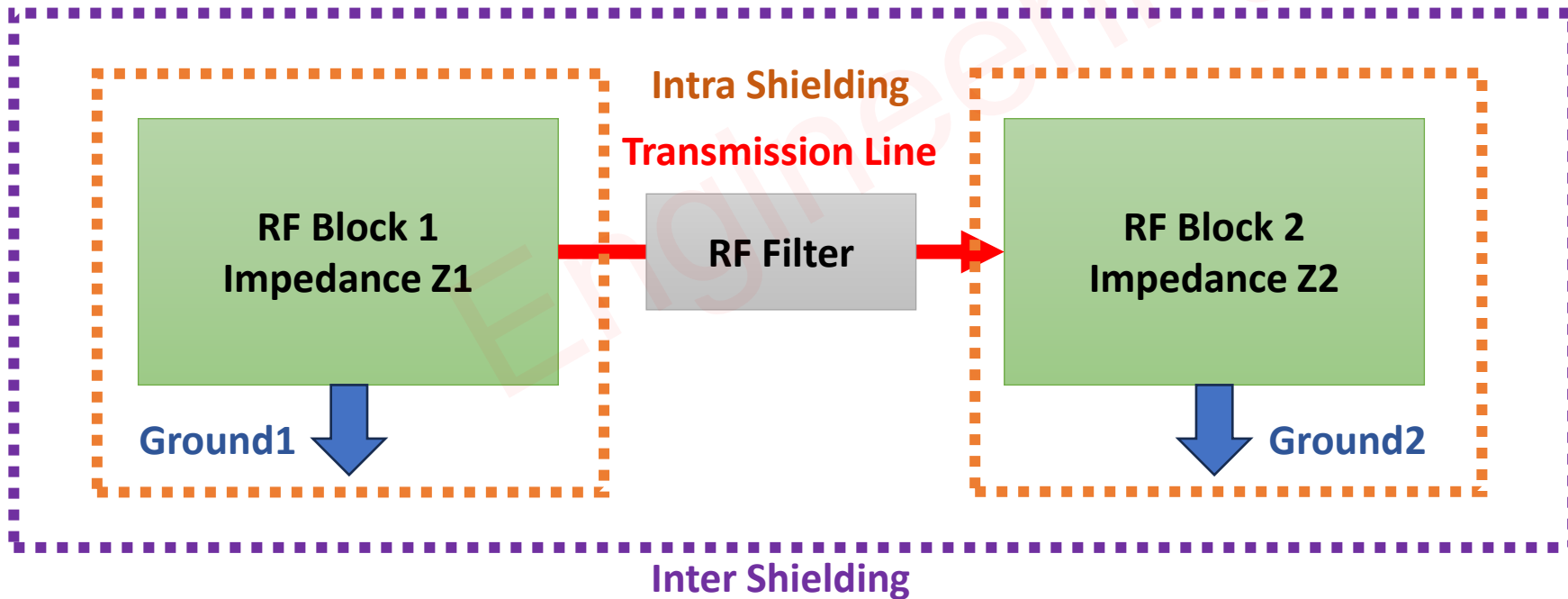
- Intra EMI: It explains the RF Interference of different blocks within the device with each other.
- Inter EMI: It explains the RF Interference of One Device with another Device.



EMC Methods

□ EMC Methods are:

- **Grounding:** Multiple RF Blocks are connected in the Device. To Reduce EMI between RF Blocks, proper grounding should be provided to Different RF Blocks of the Device.
- **Filtering:** RF Filters are used to Pass the required signals. So, using Filters, EMI can be minimized.
- **Shielding:** Shielding is used to minimize Intra EMI & Inter EMI.
- **Differential Signalling:** It is cancels EMI {In Twisted Pair Cable}.



Generation of EMI

- ❑ EMI can be generated due to many reasons, A Few reasons are discussed here:
 - Due to Mechanical vibration. {Ex. If we use a grinder at home, Low-frequency noise can be seen on TV}
 - Transmitters {Mobiles, TV Antennas, Satellite Tx Etc.}
 - Atmospheric Noise, Cosmic Noise, Solar Radiation, Lightning etc. {Natural Sources of EMI}.

Microwave Test Bench Setup

Parameters Measured by Microwave Test Bench

- ☐ Power (P)
- ☐ Frequency (F)
- ☐ Attenuation (α)
- ☐ Phase Shift (Φ)
- ☐ VSWR
- ☐ Impedance (Z)

Microwave Test Bench Setup



Klystron Supply

Isolator

Attenuator

Frequency Meter

Slotted Line

Device Under Test
{Directional Coupler}

Matched Terminator

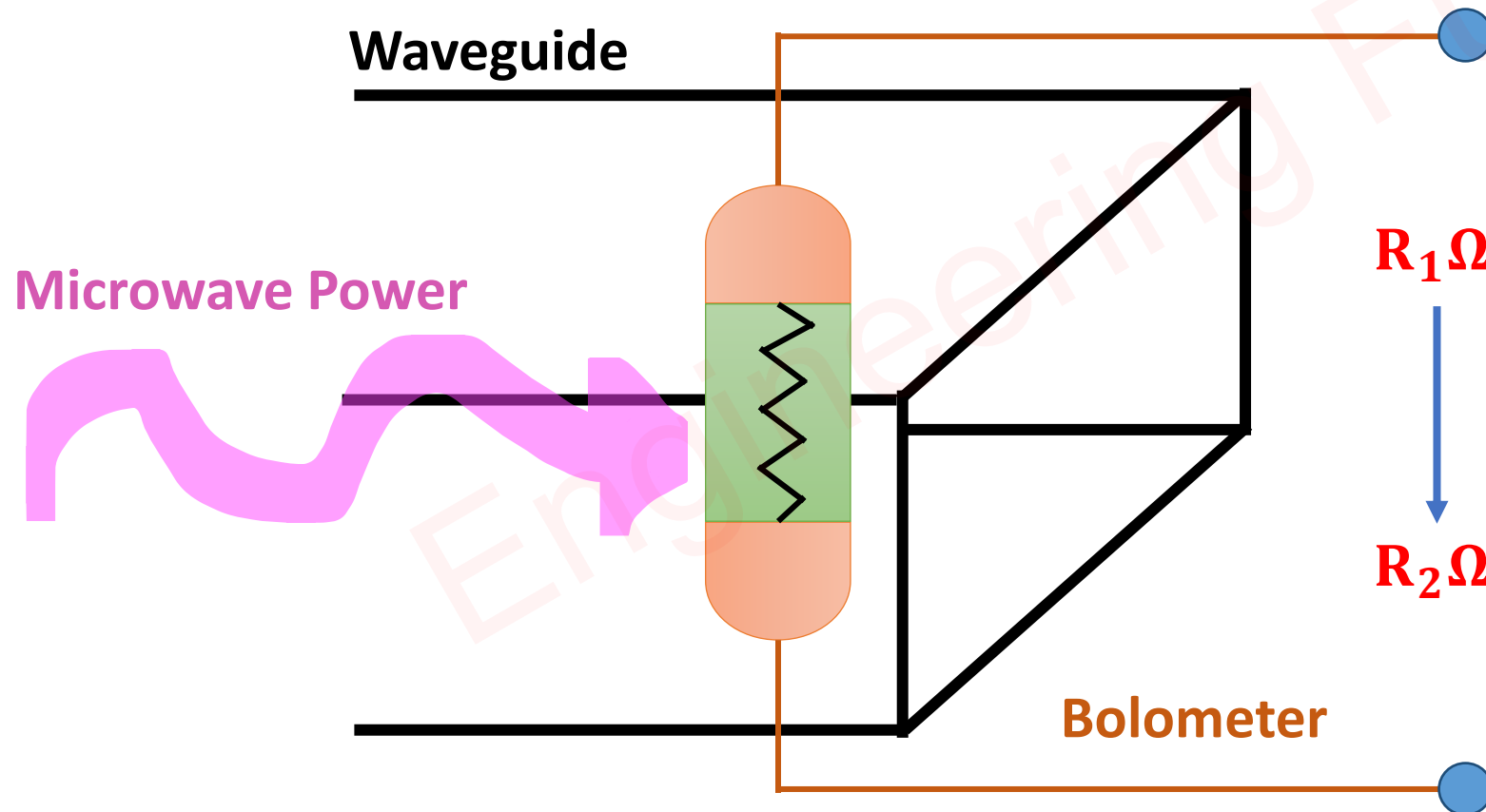
Microwave Power Measurement

Basics of Microwave Power Measurement

- ❑ Power (P) = V (Voltage) x I (Current)
- ❑ Low Power Measurement
 - Range is 0.01 mW to 10 mW.
 - It can be measured using a Bolometer.
- ❑ Medium Power Measurement
 - Range is 10 mW to 10 W.
 - It can be measured using a Calorimetric Device.
- ❑ High Power Measurement
 - Range is greater than 10 W.

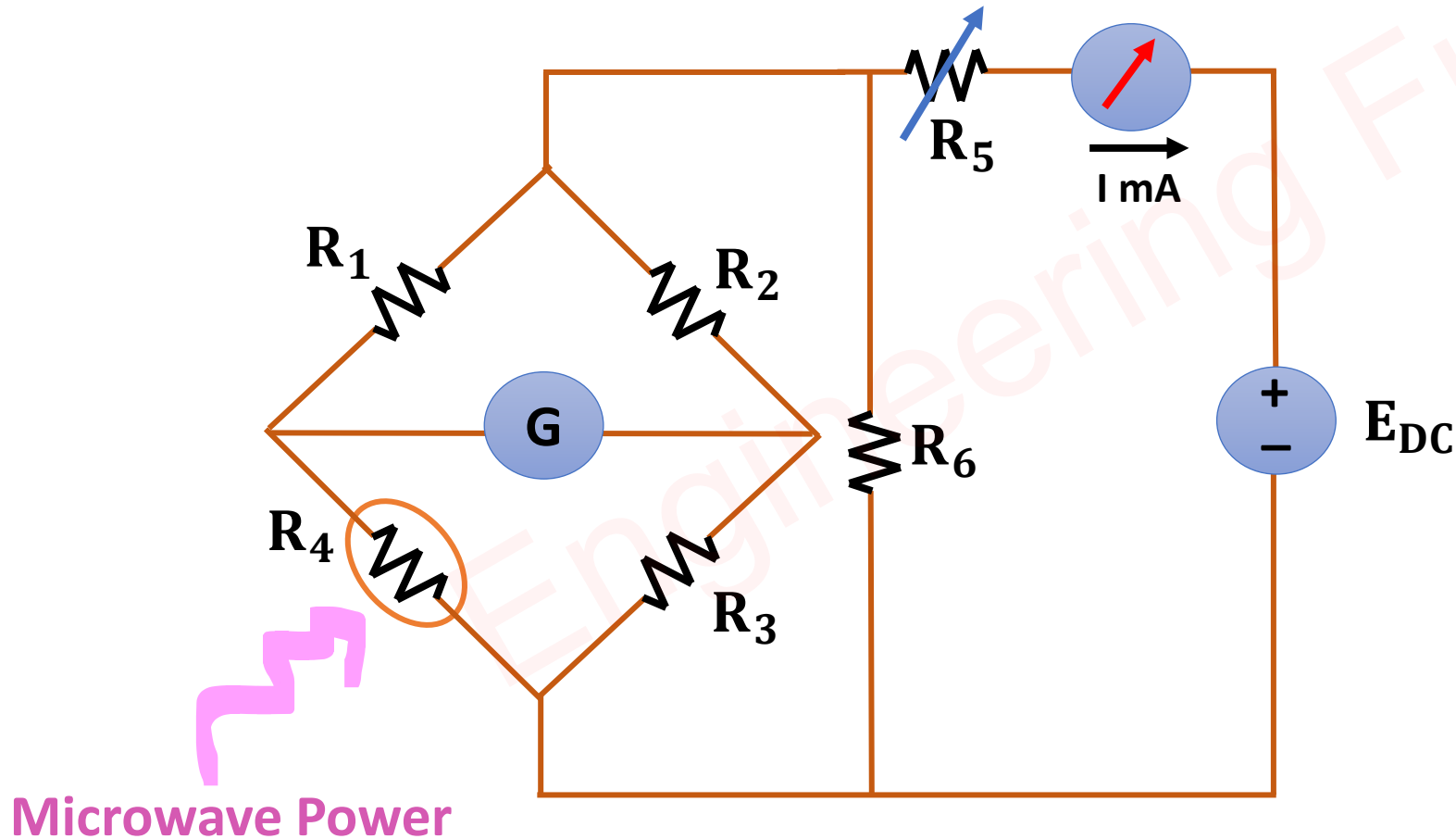
Bolometer

- ❑ Bolometer is used to measure Low Power measurement {0.01mW to 10mW}.
- ❑ Bolometer is a Resistive Type Element, The resistance of the Bolometer changes based on the EM Power on it.
- ❑ Bolometer is inserted through the slot inside the waveguide.



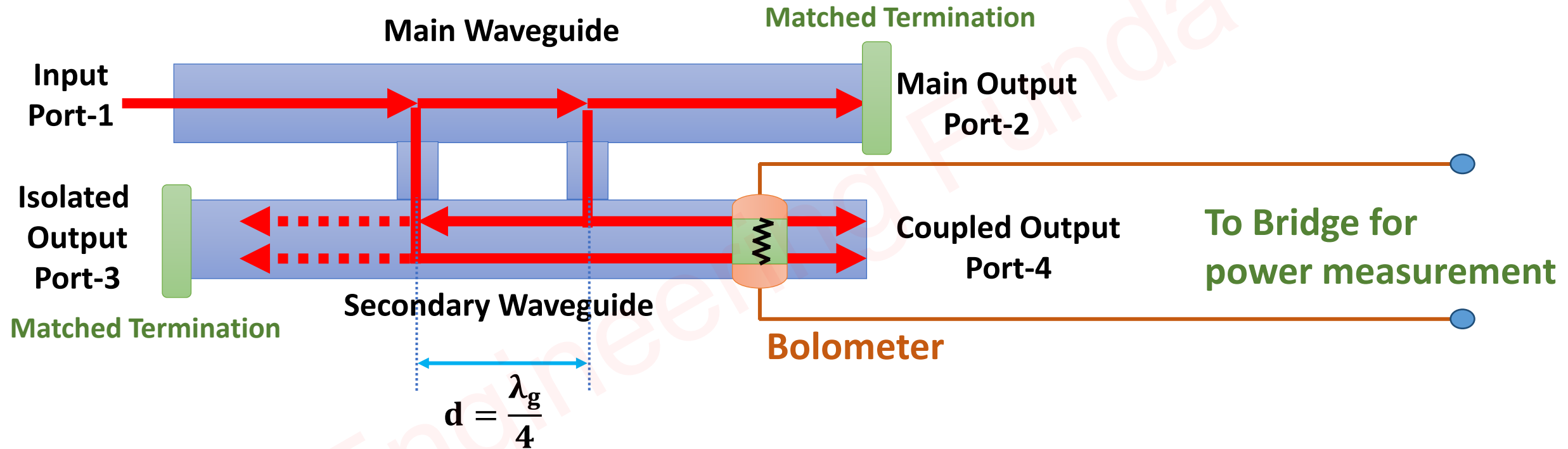
Power Measurement by Bolometer using Bridge Circuit

- ❑ Without EM-Power, Bridge is under Balanced Condition.
- ❑ After EM-Power, the Resistance of the Bolometer changes, which makes the Bridge unbalanced, Based on Current I and Voltage E_{dc} , the Power inside the waveguide is measured.



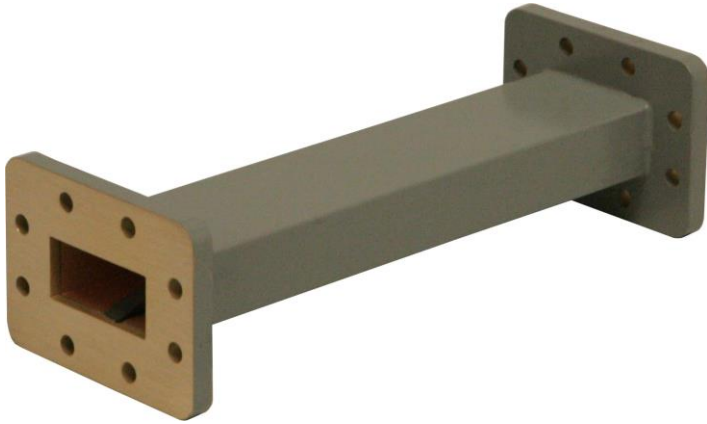
Power Measurement by Bolometer using Directional Coupler

- Directional Coupler is used to measure the higher power of EM waves, compared to simple bridge configuration.



Microwave Attenuation Measurement

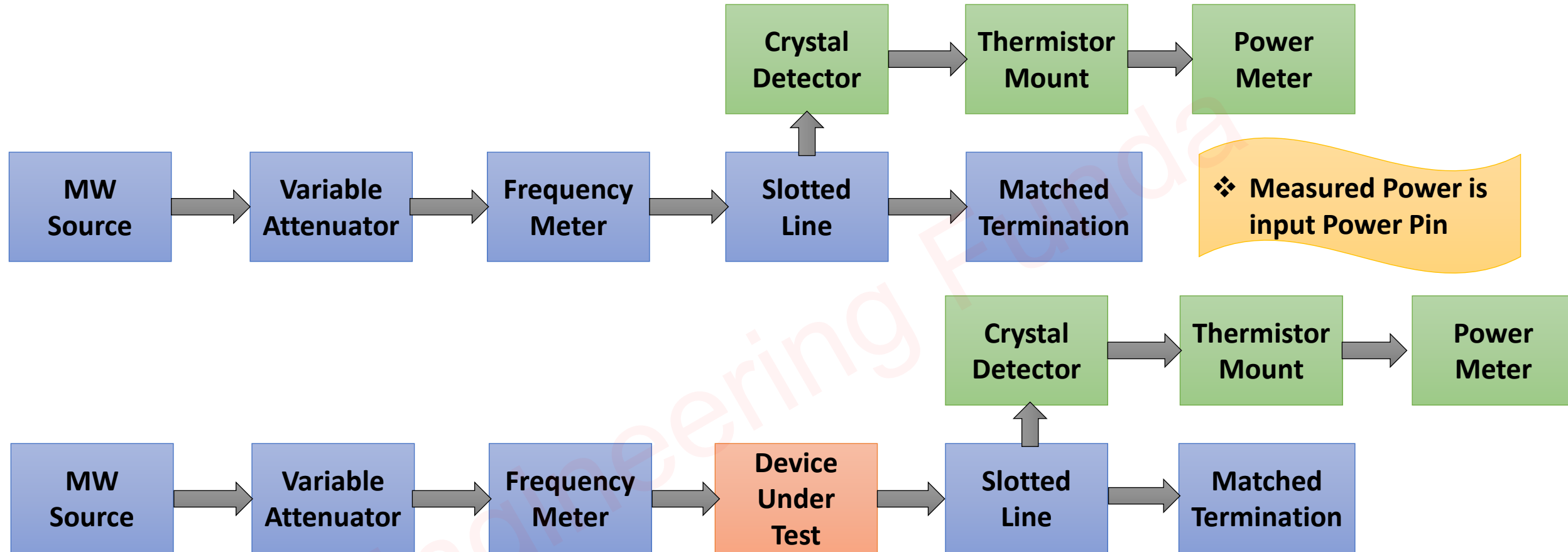
Basics of Microwave Attenuation Measurement



- $\gamma = \alpha + j\beta$
 - γ is Propagation Constant
 - α is Attenuation Constant
 - β is Phase Constant

- There are Two Types of Attenuation Measurement
 - Power Ratio Method
 - RF Substitution Method

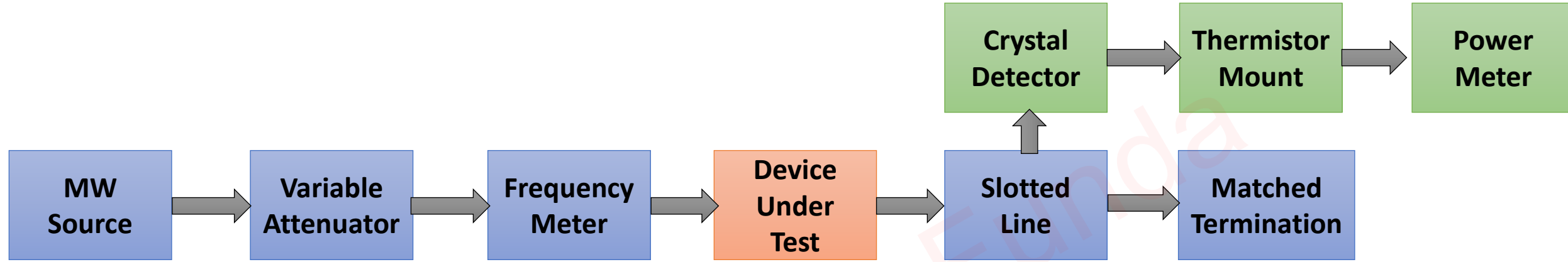
Attenuation Measurement by Power Ratio Method



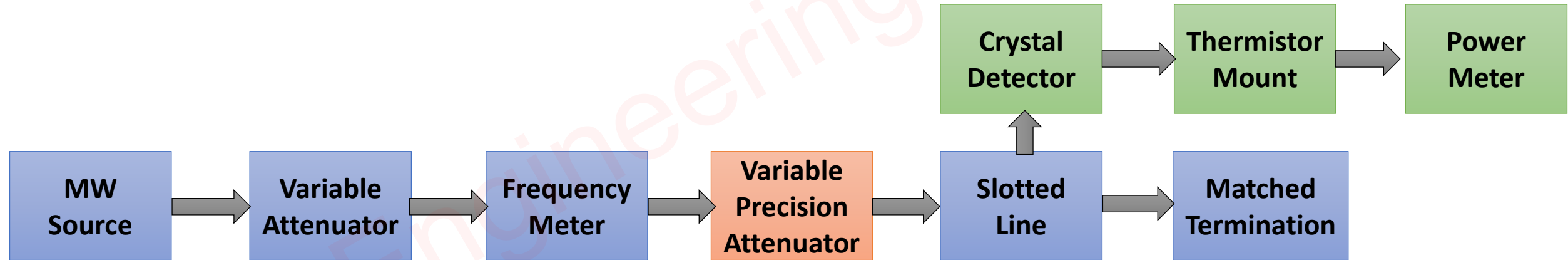
❑ Attenuation due to Device under Test is $P_{in} - P_{out}$.

❑ RF Attenuation $\alpha = 10 \log_{10} \left(\frac{P_{in}}{P_{out}} \right)$

Attenuation Measurement by RF Substitution Method



❑ Here, Lets Assume Measured Power is P .

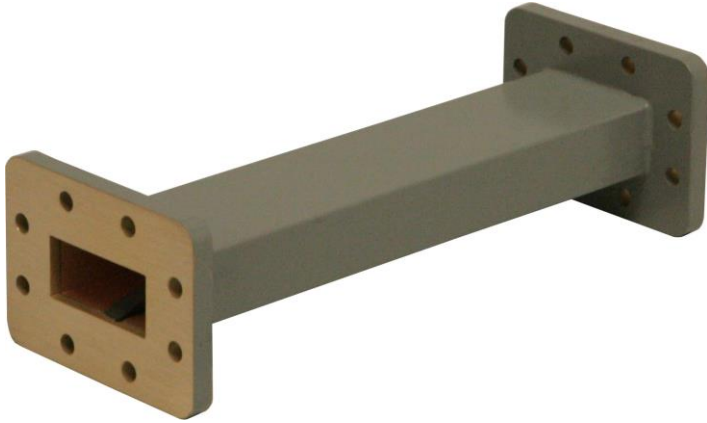


❑ Here, We change the variable Precision Attenuator until we get Measured Power equals to Power P .

❑ So, Measured Attenuation is the Attenuation shown by Variable Precision Attenuator.

Microwave Phase Shift Measurement

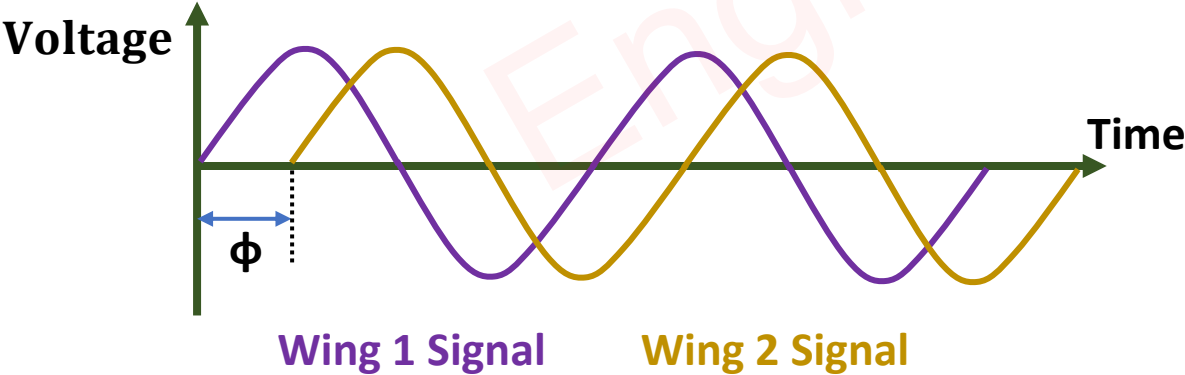
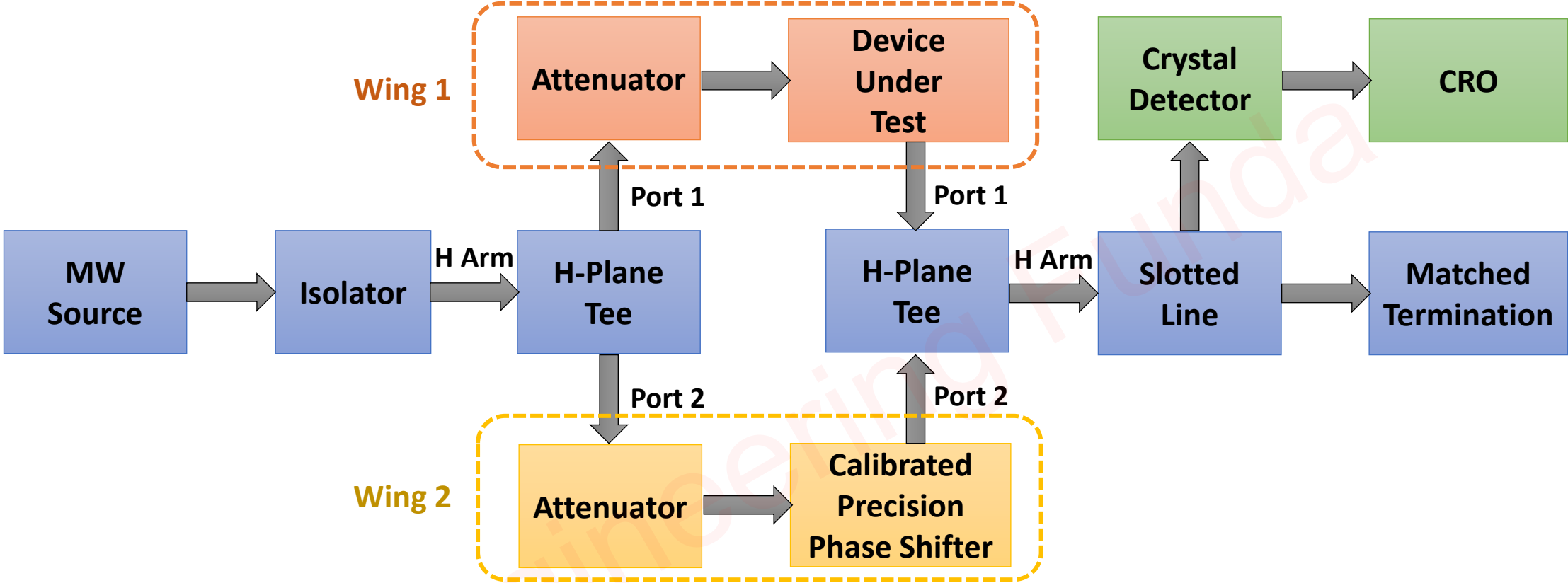
Basics of Microwave Phase Shift Measurement



- $\gamma = \alpha + j\beta$
 - γ is Propagation Constant
 - α is Attenuation Constant
 - β is Phase Constant

- Phase Constant can be expressed as $\beta = \frac{2\pi}{\lambda_g}$
 - λ_g Guided Wavelength

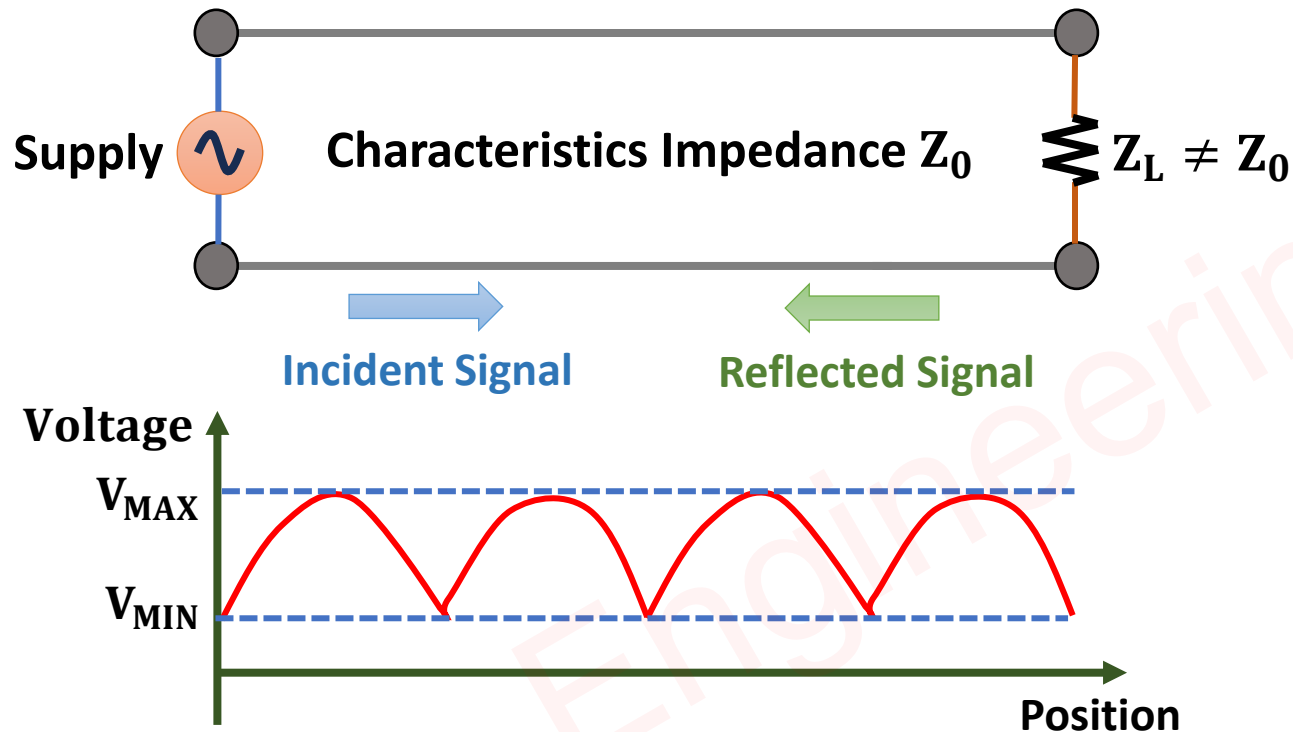
Block Diagram of Phase Shift Measurement



Microwave VSWR Measurement

Basics of VSWR Measurement

□ VSWR - Voltage Standing Wave Ratio



$$\square \text{ VSWR} = \frac{V_{MAX}}{V_{MIN}} = \frac{1+\rho}{1-\rho}$$

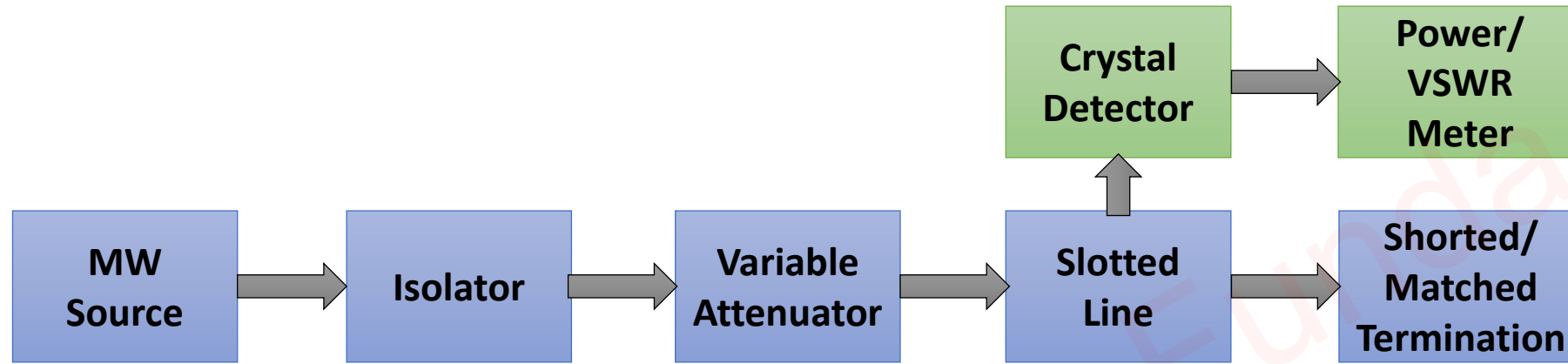
▪ Where, ρ is reflection Coefficient

$$\square \rho = \frac{V_{Ref}}{V_{inc}}$$

□ There are Two Types of VSWR Measurement

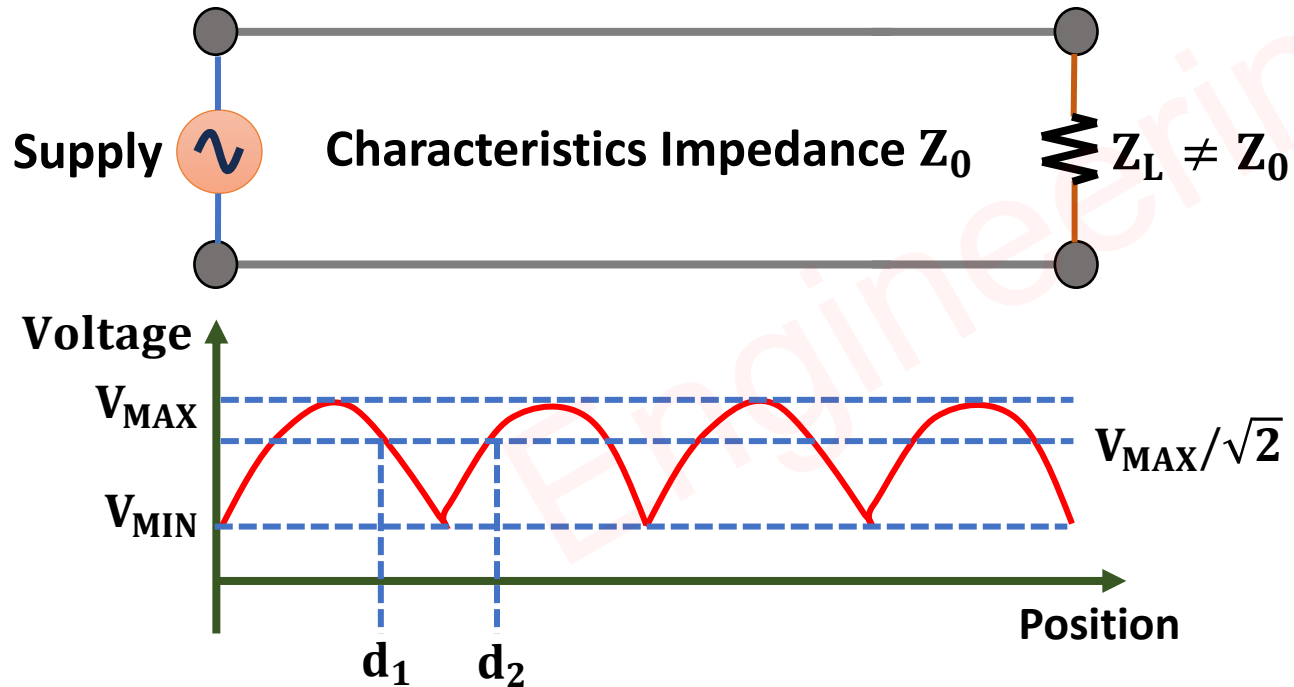
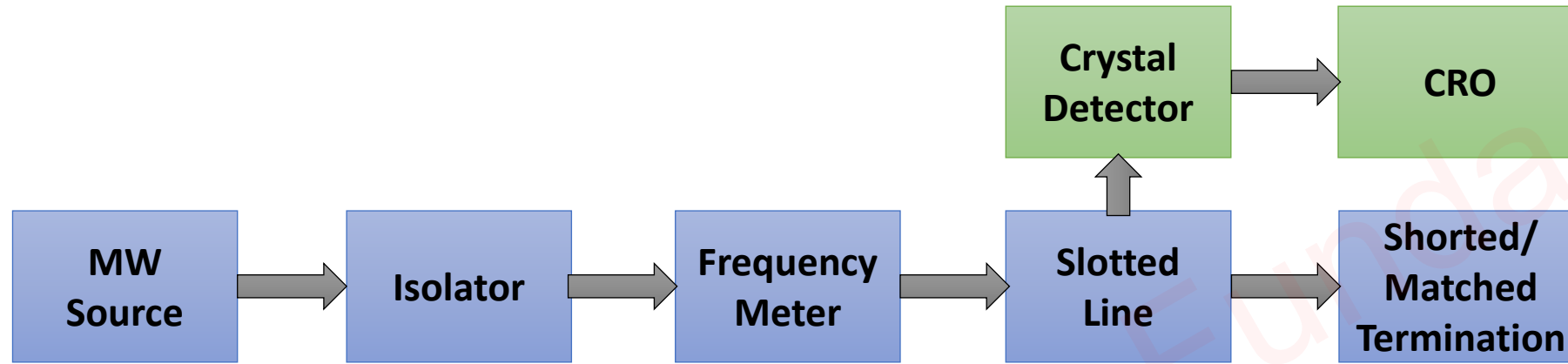
- Low VSWR Measurement ($\text{VSWR} < 10$)
- High VSWR Measurement ($\text{VSWR} > 10$)

Low VSWR Measurement ($VSWR < 10$)



- ☐ Connect the Matched Load or Shorted Load at the end of Bench Setup.
- ☐ Move the tuneable probe towards the generator until we get maximum deflection in the VSWR meter.
- ☐ At Maximum Deflection in the Power meter, we get the value of VSWR.

High VSWR Measurement (VSWR > 10)



$$\square \text{ VSWR} = \frac{\lambda_g}{\pi(d_1 - d_2)}$$

$$\square \lambda_c = 2a \text{ (for TE}_{10} \text{ Mode)}$$

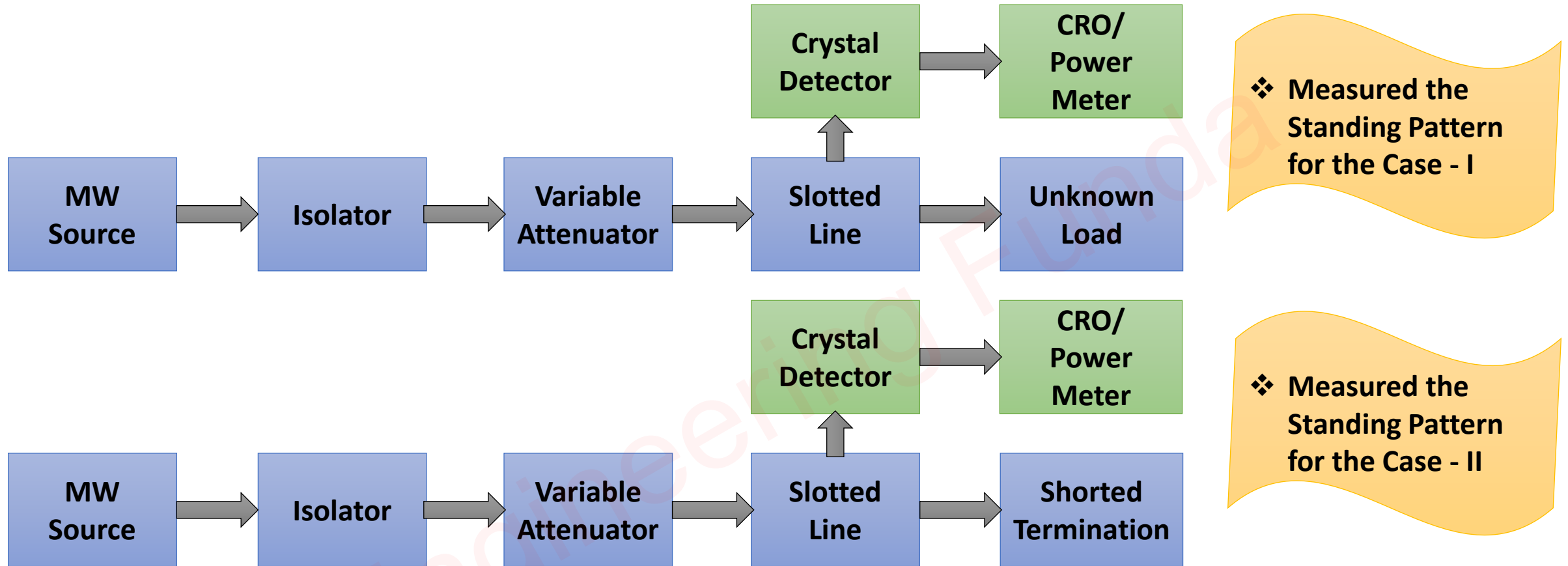
$$\square \lambda_g = \frac{\lambda_0}{\sqrt{1 - \left(\frac{\lambda_0}{\lambda_c}\right)^2}}$$

Impedance Measurement

Basics of Impedance Measurement

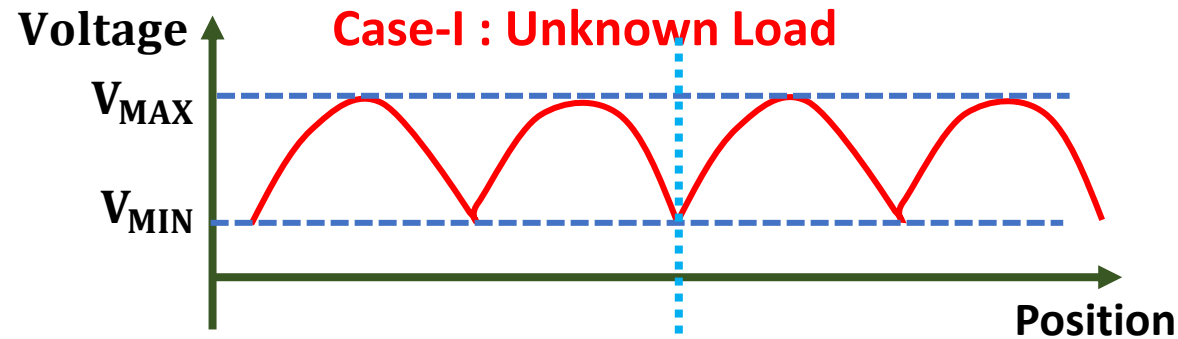
- ❑ Impedance $Z = R$ (Resistive Part) + $j X$ (Inductive or Capacitive Reactance part)
- ❑ Inductive Reactance is Positive $\{X_L = j\omega L\}$.
- ❑ Capacitive Reactance is Negative $\{X_C = -j/\omega C\}$.
- ❑ Impedance Measurement can be done by Three Different Methods:
 - Impedance Measurement by Magic Tee
 - Impedance Measurement by Slotted Line
 - Impedance Measurement by Reflectometer

Impedance Measurement by Slotted Line

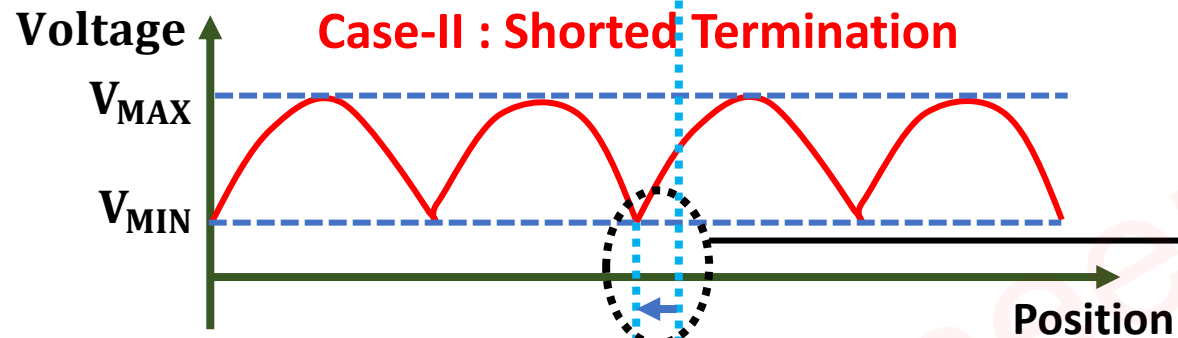


❑ Based on the Standing Pattern, we can identify the Type Load {Inductive or Capacitive}.

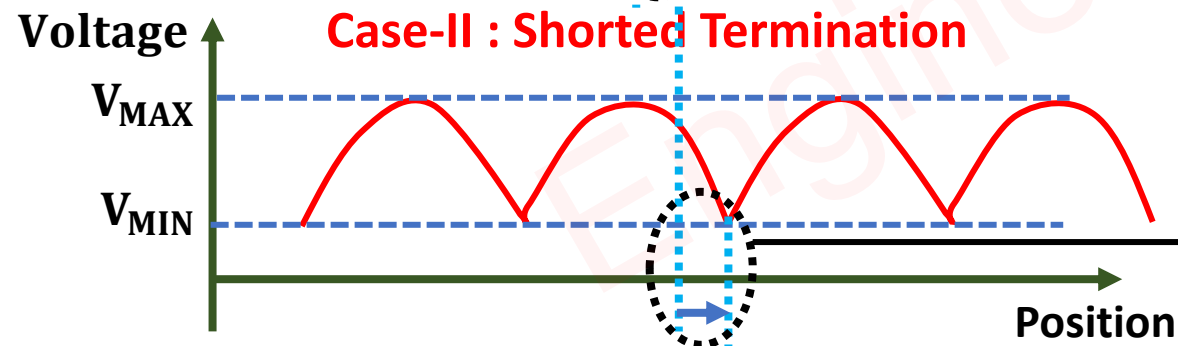
Impedance Measurement by Slotted Line



- Based on the Standing Pattern, we can identify the Type Load {Inductive or Capacitive}.
- Based on the Standing Pattern, we can identify VSWR and From VSWR we can identify the Value of Load Impedance.



Inductive Load



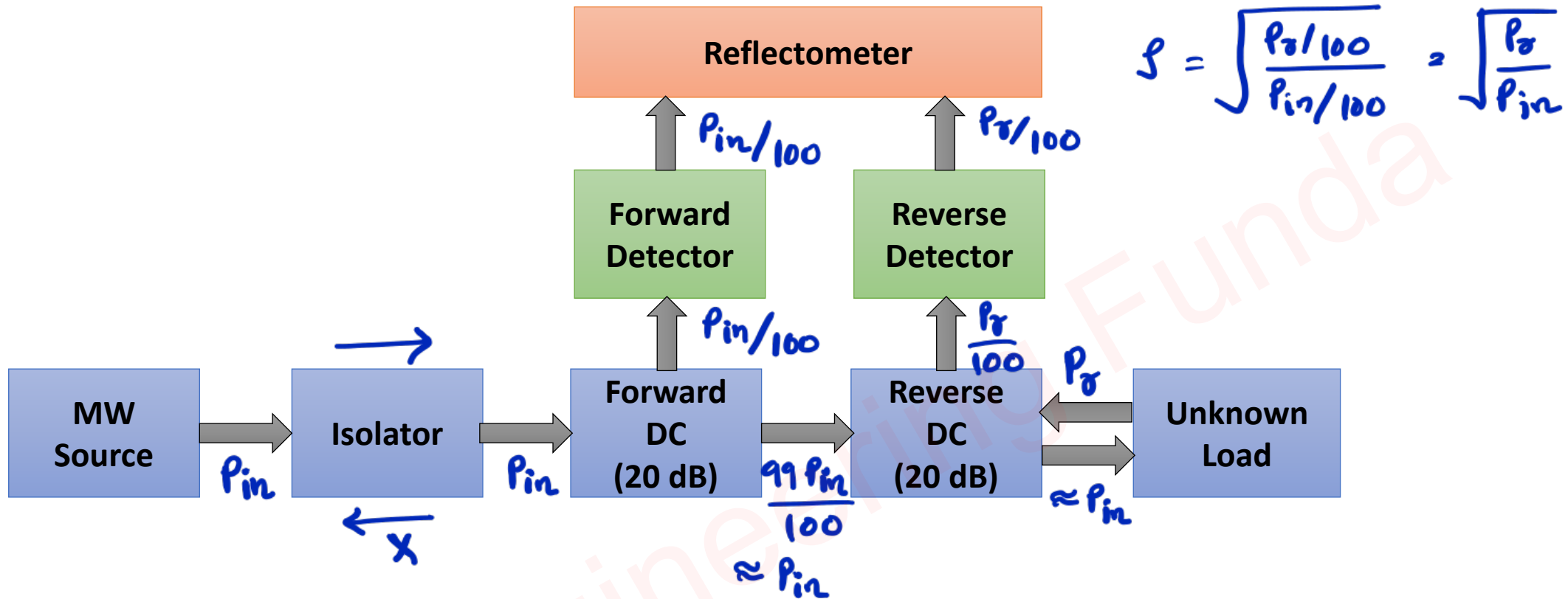
Capacitive Load

$$\Rightarrow \Gamma = \frac{V_{SWR} - 1}{V_{SWR} + 1}$$

$$\Rightarrow \Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$\Rightarrow Z_0$ is Characteristics Imp. which is known Imp.
So, Z_L can be calculated.

Impedance Measurement by Reflectometer



$$\Rightarrow \Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$\Rightarrow Z_0$ is characteristics Imp. which is known Imp. So, Z_L can be calculated.